Code No: 157EN **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech IV Year I Semester Examinations, January/February - 2023
ELECTRONIC SENSORS
(Common to EEE, CSE, IT)

Time: 3 Hours **Max.Marks:75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What is an active and passive transducer?	[2]
b)	What are the advantages and disadvantages of an electrical transducer?	[3]
c)	How is temperature range calculated using a thermocouple sensor?	[2]
d)	What are the three important aspects of a radiation thermometer?	[3]
e)	What is a Hall effect sensor?	[2]
f)	Explain Lorentz Force.	[3]
g)	Define cell constant. Mention the SI unit of cell constant.	[2]
h)	List two differences between electrochemical cell and electrolytic cell.	[3]
i)	Classify the sensors based on application.	[2]
j)	Discuss the parameters considered during selection of sensors.	[3]

PART – B

(50 Marks)

2. What is meant characterization of a sensor? Explain briefly about mechanical and thermal characterization of a sensor. [10]

OR

3. What are the semiconductor strain gauges? Explain them in detail. [10]

4.a) Describe the principle of spectroscopic thermometry.
b) Define noise thermometer. Explain various schemes of noise thermometry. [5+5]

OR

5.a) Describe Dielectric Constant and Refractive Index thermo sensors.
b) Write short notes on helium low temperature thermometer. [5+5]

6. Write a short note on Angular Movement Transducers and Semiconductor Magneto resistors. [10]

OR

7.a) Discuss the working of Synchros.
b) Explain Inductance Current Sensor. [5+5]

8.a) Discuss the working of electro ceramics in Gas media.
b) What is Liquid Junction? [5+5]

OR

9.a) Explain the Fibre optic sensors.
b) Discuss about Nuclear radiation sensors. [5+5]

10. Explain the Sensors for environmental monitoring and aerospace sensors in detail. [10]

OR

11. Describe with the help of diagram, how the primary sensors are being integrated with signal processing ensembles. [10]

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